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Weather Data Aggregator (WeatherScope)

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1. Problem Statement

The Problem:

- Weather data is scattered across platforms.
- Hard to compare temperatures between multiple cities.
- Users cannot easily calculate statistics (**average, highest, lowest**).

Requirement:

- Fetch **real-time weather data** for 5 cities.
- Compute **average, highest, and lowest temperatures**.
- Present results **visually and clearly**.



2. The Solution:

Weather Data Aggregator(WeatherScope) Dashboard:

- Fetches data via **Open-Meteo API**.
- Uses **parallel API calls** for efficiency (`Promise.all`).
- Displays **city-wise temperature cards**.
- Generates **bar chart comparison**.
- Shows **map markers** for each city.

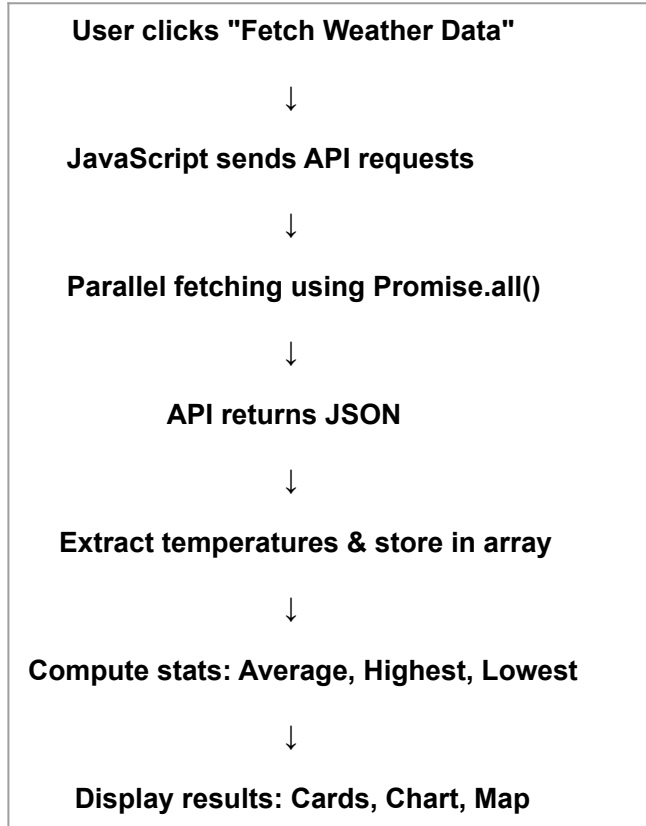
Example Output Table:

City	Temperature
Mumbai	30°C
Delhi	28°C
Kolkata	31°C
Bangalore	26°C
Chennai	29°C



3. Working

Workflow:



Aggregation Formulas:

<u>Calculation</u>	<u>Formula</u>
Average Temperature	<i>sum(temps) / number of cities</i>
Highest Temperature	<i>math.max(...temps)</i>
Lowest Temperature	<i>math.min(...temps)</i>

4.Conclusion

Outcome:

- Retrieves **real-time weather data** efficiently.
- Implements **asynchronous JS & parallel API calls**.
- Performs **statistical analysis** automatically.
- Displays results via **interactive charts and maps**.

Learning / Future Improvements:

- Add **more cities** or live news integration.
- Include **forecast predictions**.
- Improve **visual presentation and responsiveness**.





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Thank You.